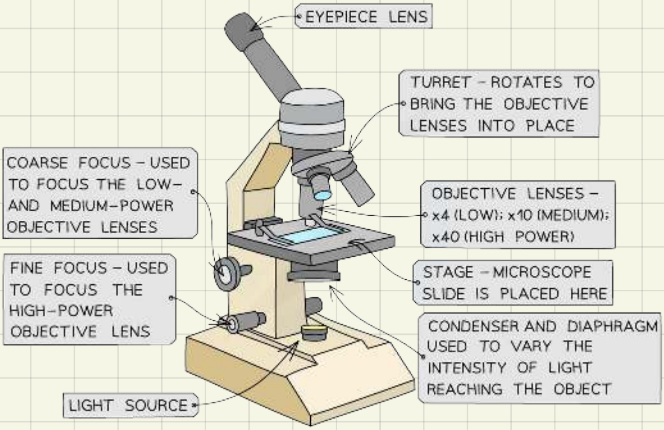
Microscopes:

① light microscope

- uses a source of light [Sun/ Electricity]
- eyepiece lens and objective len

*To find magnification of microscope multiple mag of objective lens by eyepiece lens

- stage to place specimen
- condenser



- **Advantages:**
  Relatively Cheap
  Relatively light and portable
  Can see through living tissues

- **Disadvantages:**
  Limited magnification and resolution
  Preserving and staining can produce artifacts which can be mistaken for a tissue part

$$\text{Actual size} = \frac{\text{image size}}{\text{magnification}}$$

Types of stains:

1) haematixylin —> stains animal and plant nuclei purple blue or brown
2) methylene blue —> stains animal nuclei blue
3) acetocarmine —> stains plant and animal chromosomes in dividing nuclei
4) iodine —> stains starch containing material in plant cells blue-black

② electron microscope

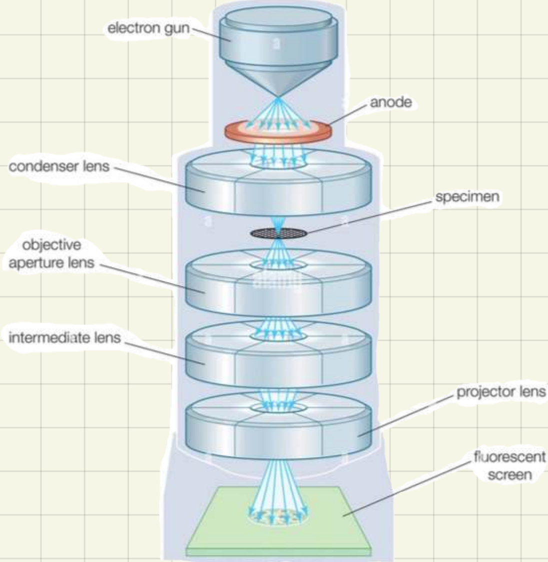
- uses a beam of electrons with short wavelength
- allows smaller structures to be viewed in greater detail

- **Advantages:**

High resolution and magnification

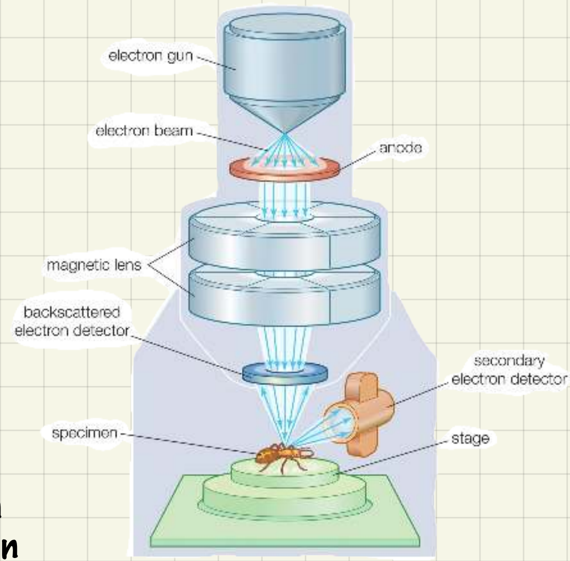
- **Disadvantages:**

Expensive
Not portable, must be kept under certain conditions, constant temperature, pressure and internal vacuum
All specimens should be examined in vacuum, impossible to look at living material
Specimen might contain artifacts due to staining



Transmission em (2D)

* heavy stains are used, such as:
Lead and uranium.

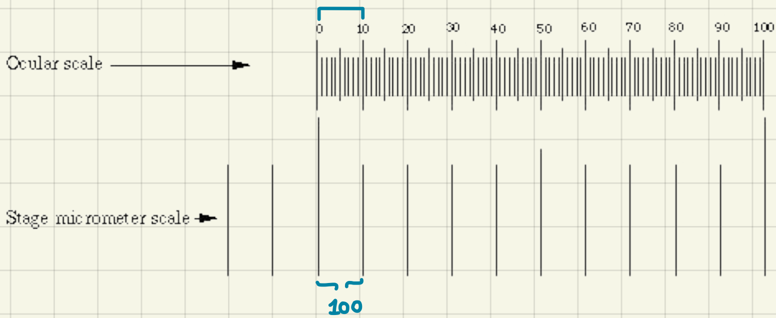


Scanning em (3D)

* low power = low magnification
* high power = high magnification

How to find the width/length of a structure?

1) Calibrate the eyepiece graticule
2) Place a micrometer slide on the stage of the microscope.
The smallest division of the micrometer scale is usually 100 µm.
3) Align the scales of the eyepiece graticule and the stage micrometer in the field of view.
4) Count the number of divisions (eyepiece units or epu.) on the eyepiece graticule that are equivalent to a known length on the micrometer slide
• work out the length of each eyepiece unit.
For example, if 100 µm is equivalent to 10 epu, then each epu is = 10 µm at this magnification



What is meant by the terms,

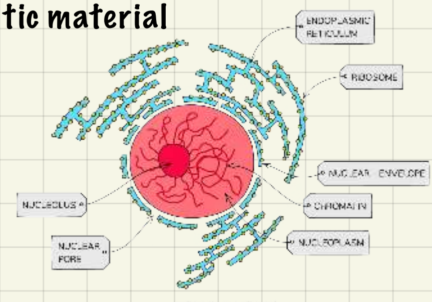
Resolution:
a measure of how close 2 objects must be before they are seen as one

Magnification:
a measure of how much bigger the image you see is than the real object

Artifacts:
things observed in a scientific investigation that are not naturally present, occurs as a result of the preparation

Eukaryotic cells

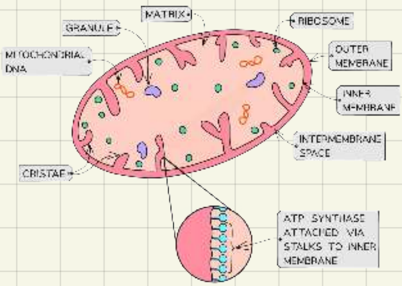
- **protoplasm**: cytoplasm + nucleus
- **Nucleus**: largest structure, contains genetic material
  Double membrane enveloping
  nuclear pores



Ribosome synthesis happens in nucleolus

* Chromatin = DNA + protein

- **Mitochondria**: site of cellular respiration to produce ATP, folded double membrane for increased surface area for aerobic respiration



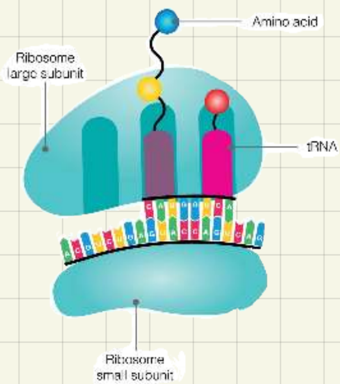
*matrix is a gel like fluid

*Has 70s ribosome

*Has its own genetic material, plasmids

- **lysosomes**: small, dark and spherical, contain enzyme lysosame.
  Digests dead cells/ waste material

- **ribosomes**: ribosomal RNA + proteins
  —> For protein synthesis



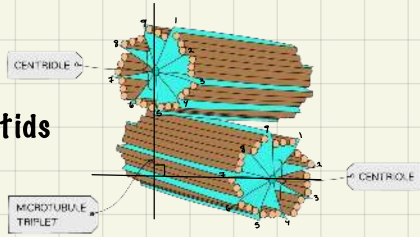
Types:

small ribosome	Large ribosomes
70s	80s

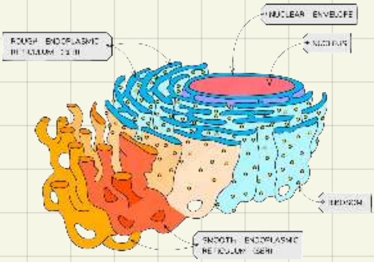
ratio of small to large subunits in each type:

40:60	30:50

- **Centrioles**: cylindrical shape, each centriole is made up of 9 tubules (arranged at right angles)
  - * only found in animal cells
  - —> provides spindle fibers
  - —> mitosis to separate chromatids



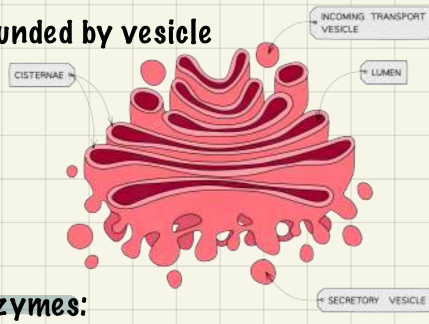
- **Endoplasmic reticulum**: sac interconnected with nucleus



- Types:
  1) rough ER → has ribosome for protein synthesis and protein transport
  2) smooth ER → no ribsome, synthesizer of lipids

Has large surface area for synthesis, transport, storage of proteins.

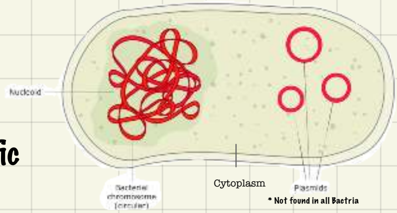
- **Golgi body**: flattened and curved sacs, surrounded by vesicle (arranged up to size)
  - —> for protein modification
  - [protein → lipoprotein]



*rER and Golgi body in releasing extracellar enzymes:

- Protein is synthesized on ribosome.
- Protein enters ER where it is folded into tertiary (3D) structure
- Protein moves through ER and is packaged at one end then is pinched out into a vesicle
- Vesicle moves to Golgi body and fuses with it
- Protein is secreted into golgibody and is modified [such as adding carbohydrates forming glycoproteins]
- Modified protein is pinched out in a secretary vesicle which moves to cell surface membrane and fuse with it, releasing extra-cellular enzymes by exocytosis

Prokaryotic cell

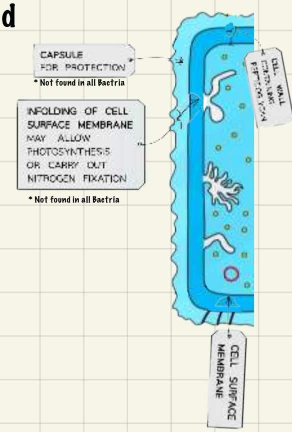


- **Nucleoid:** area where genetic material is found
  → single circular strand of DNA

- **Plasmid:** extra genetic material coding for specific bacterial phenotype

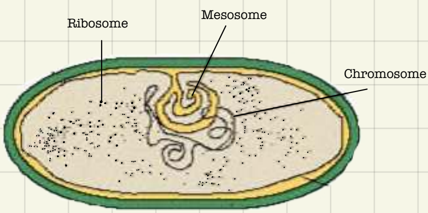
- **Cytoplasm:** site of metabolic reactions

- **Cell membrane:** partially permeable bilayer of phospholipid



- **Cell wall:** made from peptidoglycan
  → prevents swelling and bursting
  → maintains bacterium shape
  → for protection and support

- **Capsid:** protection against phagocytosis (by covering cell markers on membrane that identify cell)
  * not all bacteria have capsid usually found in Bactria



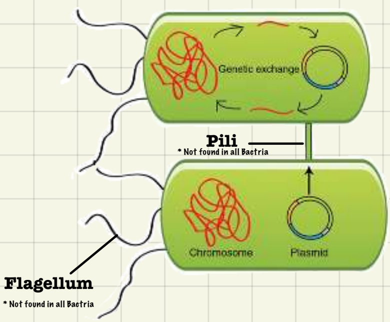
- **70s ribosomes:** for protein synthesis

- **Mesosome:** provides energy

- **Pili:** protein projections
  → attachment to host cell
  → sexual reproduction

*can make Bactria more vulnerable to virus infection as it's used as entry point

- **Flagellum:** stranded helix of flagellin protein
  → for movement



Congestion

— Classification of prokaryotes

① structure of cell wall:

take a sample and dye it with gram stain (is originally violet)

a) if sample appears violet after staining → gram positive bacteria

b) if sample appears pink/red after staining → gram negative bactria

② shape and arrangements

a) shape

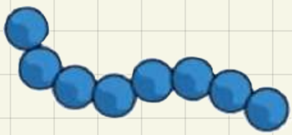
Cocci

Bacilli

Vibrio

Spirillum

b) arrangement



③ respiration type:

a) obligate aerobic (requires oxygen)

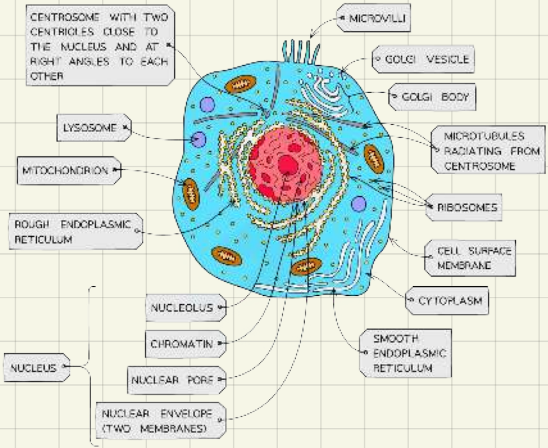
b) obligate anaerobic (requires absence of oxygen)

c) facultative (can survive in both)

Organelle —> Cell —> Tissue —> Organ —> Organ system —> Organism
(Nucleus). (Xylem). (Leave). (Digestive). (Human)

* tissue: group of specialised cells carrying the same function
* organ: different types of tissues carrying a particular function
* organ system: group of organs working together to preform a specific function

Eukaryotic cells VS. Prokaryotic cells



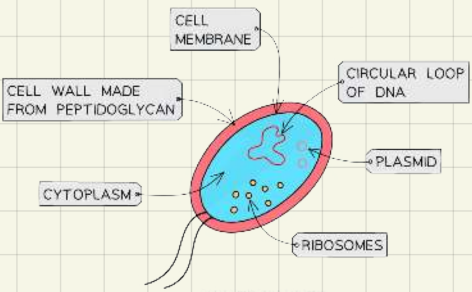
Both have cell membrane

Both have 70s ribosomes

Both prokaryotic and eukaryotic plant cells have cell wall

Both prokaryotic and eukaryotic sperm cells have flagella

PROKARYOTIC CELL

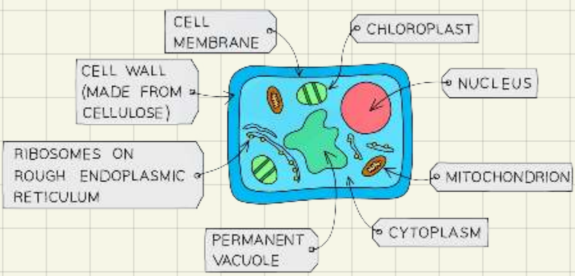


Eukaryotic cells have membrane bounded organelles while prokaryotic cells don't

Eukaryotic cells contain nucleus while prokaryotic cells don't

Eukaryotic cells have both 70s and 80s ribosomes while prokaryotic cells only have 70s ribosomes

TYPICAL PLANT CELL



Prokaryotic cells have cell wall while eukaryotic animal don't

Prokaryotic cell wall is made from peptidoglycan while eukaryotic plant cell wall is made from cellulose

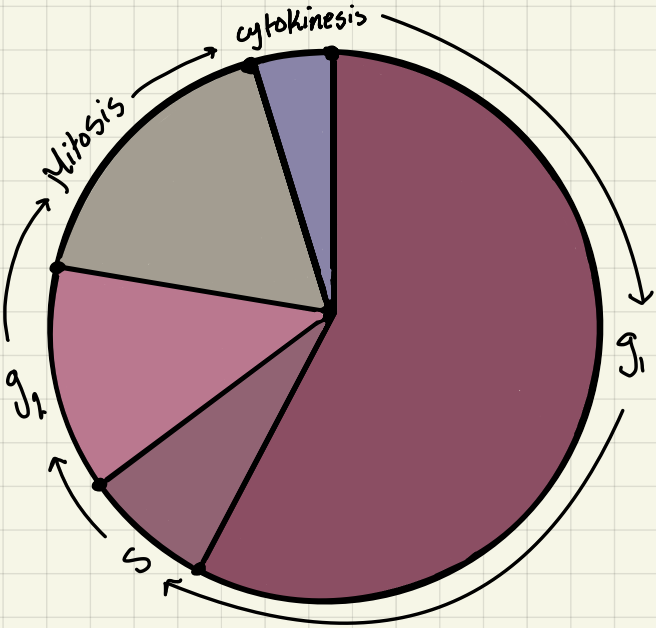
The cell cycle

Chromosome: double strand of DNA wrapped around histone protein
Locus: the location of a gene on a chromosome
Chromatin: unwrapped genetic material
Chromatid: condensed replicated DNA molecule (connecting at centromere)

Chromatin → chromosome → chromatid

* Chromosome can't be seen under light or electron microscope, only during cell division when chromosomes condense and shorten they take up enough stain to be seen

The cell cycle is controlled by small protein called cyclin which build up and attach to kinase enzyme which is found at different parts of cell causing phospho-relation and bringing on the next stage in cell cycle



Advantages of mitosis	Disadvantages for mitosis
For growth and repair	Less genetic variation
Asexual reproduction	Less adaptation to new environment
No need to find a mate	
Produces large numbers quickly	

Cell cycle:

① **interphase.**

Cell preparation for cell division
All cellular activities are carried out normally
Longest phase, may vary in length from one cell to another

Stages:

G1 Growth of cell increase in mass and size
S DNA replication
G2 Synthesis of organelle

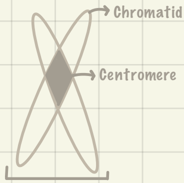
② **mitosis**

Production of genetically identical daughter cells

Stages

Prophase

Chromosome coil and shorten, each chromosome is made up from 2 daughter chromatid attached to each other at centromere
The nucleolus and nucleus membrane break down
Centrioles move to opposing poles



Metaphase

spindle fibers form
The centromere of each chromatid is connected to both spindle fibers but each centromere is alone at spindle microtubule
Chromatid are aligned in the equator

Anaphase

Spindle fibre contracts as a result chromatid separated
Sister chromatids are pulled to opposite poles
Each chromatid is now called chromosome
This is an energy-using process

Telophase

Spindle fibre breaks down
Nuclear membrane forms
Nucleoli and centrioles also reform
Chromosomes unravel and become less dense

③ **Cytokinesis**

Cytoplasm divides
In animals this happens by contractile fibers tightening around cell center until the two cells are separated
In plants this happens by a cellulose cell wall being build up from center to outside

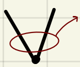
- **Mitotic index** = $\frac{\text{no. of cells undergoing mitosis}}{\text{total no. of cells}}$

* only cells undergoing mitosis will have visible chromosomes
* used to see how active cell development is
(cancer patients have high mitotic index)

- **The Role of cell cycle:**
  Synthesis of organelles
  Increase in size (growth of cell)
  Synthesis/ replication of DNA

- **Root tip squash** (to study mitosis)

1) take tissue (2 - 5mm from onion root tissue)



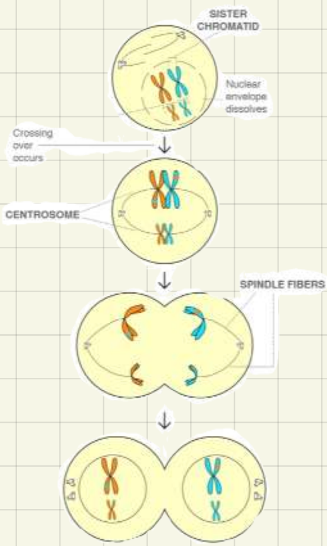
2) put tissue in HCL acid to soften the tissue
3) stain it with acetocarmine
4) place tissue on slide then put cover slip and squash ,to flatten tissue, gently, to not damage cells.
5) add more stain drops
6) warm it by hovering over Bunsen burner to intensify stain color

* start with microscope lowest magnification power
* count number of cells undergoing mitosis (visible chromosome) and the total number of cells, calculate mitotic index

Meiosis

Reduction division for development of gametes

① meiosis (1)



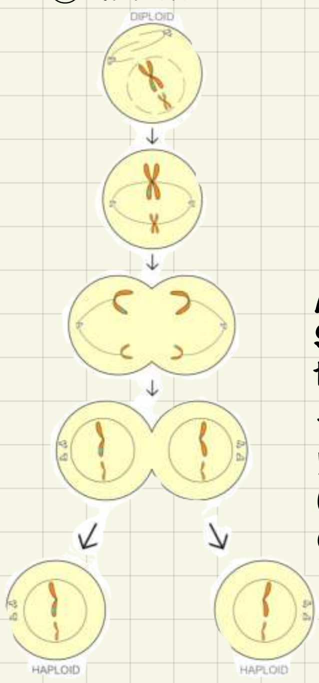
Prophase 1
DNA is replicated in interphase so condensed chromosomes form 2 chromatids
Homologous pairs associate with each other.
Crossing over occurs

Metaphase 1
Each centromere is attached to one spindle fibre
Chromatid are aligned at equator

Anaphase 1
Spindle fibre contract pulling whole chromosome from each homologous pair to opposite poles

Telophase 1
Nuclear membrane reform and cells being to divide

② meiosis (2)



Prophase 2
DNA replication doesn't occur before prophase 2
Nuclear membrane disappears, new spindle forms

Metaphase 2
Each chromatid pair is attached to 2 spindle fibre at centromere; align in equator

Anaphase 2
Spindle fibre contract, centromere divides and the chromatids move to opposite poles

Telophase 2
Nuclear membrane reforms
Cell split by cytokinesis forming 4 haploid daughter cells

- Importance of meiosis:

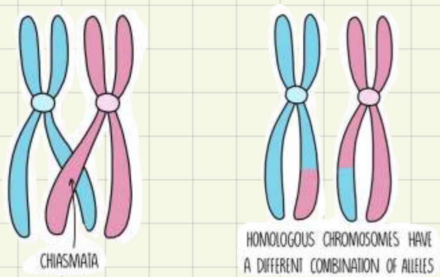
Production of haploid gametes to maintain original number of chromosomes after fertilization
Genetic variation
More adaptable to environmental changes

- Disadvantages of meiosis:

Slow
Needs to find a mate

- genetic variation is introduced by:

A) crossing over: Occurs in prophase 1
when the homologous chromosomes move towards each other and exchange genetic material.
[alleles on same locus get interchanged]



B) independent assortment: occurs in metaphase one
Depends on the order in which chromosomes line up along the equator of the cell during metaphase different combinations of chromosomes will end up in each gamete.
*Is completely random so many different combinations can occur

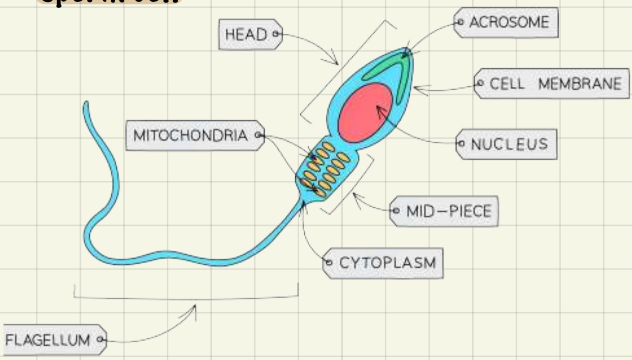
Sexual reproduction

Gametes: haploid cells which fuse together to make a diploid zygote

In animals:

	Male	Female
Gonad:	Testes	Ovary
Gamete:	sperm (microspores)	Ovum (megaspore)

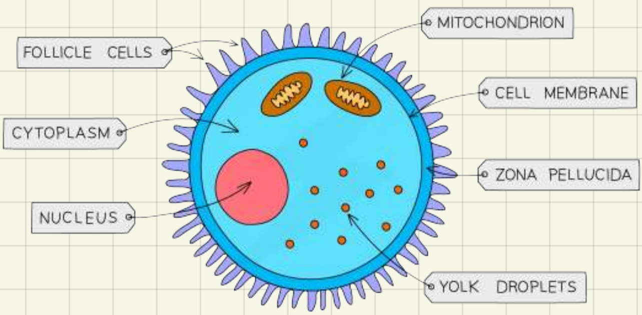
Sperm cell



-Nucleus with haploid chromosomes to maintain original no. Of chromosomes after fertilization
-Acrosome contains enzyme to penetrate ovum protective barrier
-Mitochondria to produce energy for movement
Flagellum for movement to ovum

Ovum

-Nucleus with haploid chromosomes to maintain original no. Of chromosomes after fertilization
-Protective jelly-like layer (Zona pellucida)
-Large as it has food store for zygote



Fertilization in humans

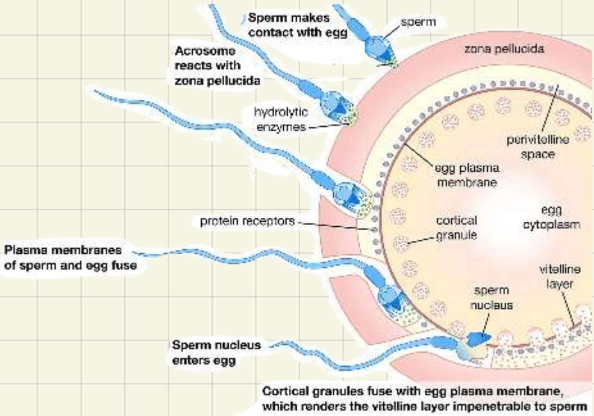
Internal fertilization: where male gametes are transferred directly to female.

Ovum + sperm = zygote
23 + 23 = 46

1) acrosome reaction
Fuse between sperm head and acrosome. Then enzyme is secreted by exocytosis to digest zona pellucida
* one acrosome is not enough to digest all zona pellucida
So multiple sperms release their enzymes

2) only one haploid nucleus enters the ovum to fuse with female haploid nucleus to form zygote

3) cortical reaction
Cortical granules shift to the outside and harden to form fertilization membrane preventing polyspermy
* cortical reaction takes time
So membrane gates open and close making a once positive ovum negative repealing negative sperms



Zygote then multiples by mitosis and nutrients needed are provided by ovum up until it is developed into an embryo which is moved from oviduct to uterus and then placenta forms so nutrients are provided through it

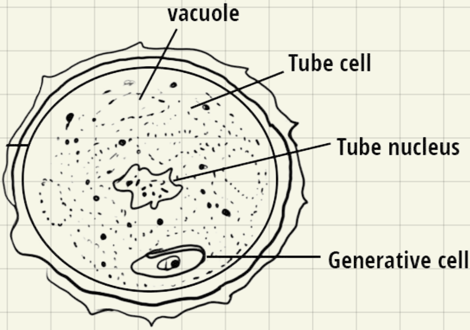
*Fertilization in humans is called conception

In plants:

	Male	Female
Gonad:	Anther	Ovary
Gamete:	Inside Pollen	Ovule

*Fertilization occurs internally

Microspore in flowering plants



* cells are produced by meiosis but no cytokinesis so both haploid nuclei stay in cell

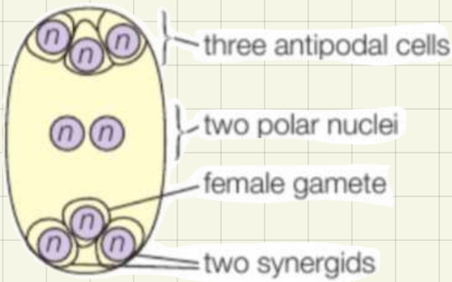
- generative nucleus

fertilizes ovum to produce zygote

- tube nucleus

penetration of stigma, style and ovary by forming pollen tube

Magaspore in flowering plants



* 2 meiosis occur then 3 mitosis nut no cytokinesis producing 8 haploid nuclei

- female gamete

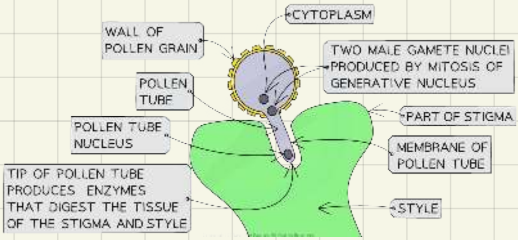
fuses with generative nucleus to form zygote

- two polar bodies

fuse with second generative nucleus to provide food for zygote

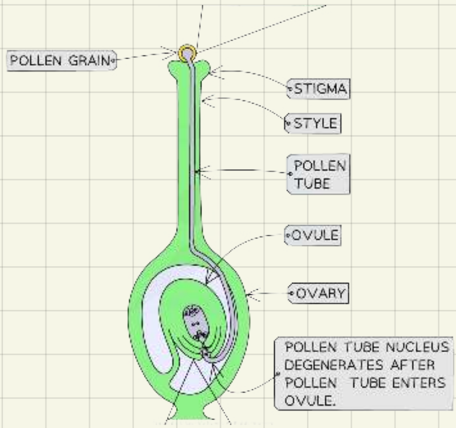
Fertilization in plants

Fertilization only occurs ① when stigma is mature
② pollen grain and ovule are from same species



1) pollination
Pollen grain lands on stigma and if both are from same species but different plants pollen grain will germinate

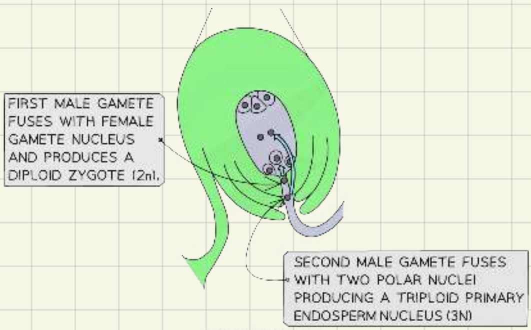
2) tube nucleus will release hydrolytic enzymes to digest style creating a pollen tube which ends at micropyle



3) as generative nucleus enters pollen tube it divides by mitosis to produce genetically identical nuclei

4) double fertilization
First male nucleus to reach egg cell forms zygote
Second nucleus fuses with the two polar bodies to form endosperm (triploid nucleus) which supplies embryo with nutrients

*Ovary develops into fruit



* external fertilization is when female and male gametes meet and fuse in external environment mostly occurs in aquatic organisms

Cell differentiation

All cells contain the same genetic material yet each cell express the genes they need to carry on their function

All cells express "housekeeping" genes which produce basic cell proteins such as structural membrane proteins and enzymes needed for respiration

- Transcription factors

  Are proteins which have binding site to :

1) promoter sequences (specific regions on DNA)

Found at the top of a transcription upstream of a gene a
They stimulate the start of the transcription process

2) enhancer sequences (specific regions on DNA)

Changes the structure of Chromatin by making it more

→ Open to RNA polymerase
→ Closed to RNA polymerase

<u>More open → active gene (switched on)</u>

<u>Less open → non- active gene (switched off)</u>

* one gene can bind to multiple transcription factors * one transcription factor can bind to multiple genes

Transcription factors can produce different types of cells by switching the genes on and off during different development stages and body circumstances.

- Non-coding RNA (ncRNA)

Much of human genome doesn't code for proteins
Only around 2% of RNA codes for protein the rest are called ncRNA
One X chromosome is inactivated at random in each nucleus to form a supercoiled, condensed stable Barr body

* unique non-coding DNA regions are called <u>introns</u>

How do genes get specialized?

-up to chemical stimuli
-some genes get switched on (active)
-some get switched off (not active)
-through epigenetic modifications
-only the switched on (active) genes get transcribed and translated into specific proteins.
-These specific proteins modify cell structure and function

Epigenetic modifications:

1) RNA splicing (post-transcriptional modification)

Pre-RNA → RNA before modification is still made up from both introns and exons (codes for protein, is the same in all individuals)

Mature RNA → Where all introns and exons are split, hydrogen bonds between complementary bases (exons and introns) break
Only Needed exons are joined together by the enzyme spliceosome
* mature RNA is shorter than pre-RNA

Mature RNA is then transcribed to mRNA

Spliceosome enzyme can join exons in different combinations producing different mRNA versions from the same section of DNA as a result each version of mRNA codes for a different arrangement of amino acids so different polypeptide are made

Why can one gene code for more than one protein ?

Up to post-transcribitional modification spliceosome enzyme cuts pre-rna removing introns from exons. Splicesome enzyme joins the same exons in different orders as a result a single gene can produces several versions of functional mRNA. Then functional mRNA is translated to different types of proteins

2) DNA modification

- <u>DNA methylation</u>

Addition of methyl group between adjacent cytosine and guanine
It silences a gene or a sequence of genes so they can't be transcribed nor translated

-DNA demethylation is removal of methyl group which makes the gene/ sequence of genes active so they are transcribed and translated to produce proteins

3) Histone modification

- <u>histone acetylation</u>

Addition of acetyl group to a lysine (amino acid) in histone (protein) Which opens up (loosens) the structure and activate the Chromatin.
Genes in that area can get transcribed and translated into proteins

Removing acetyl group silences the gene area again forming heterochromatin (condensed and supercoiled with unavailable gene areas)

- <u>histone methylation</u>

addition of methyl group to lysine in histone
Depending on the position of lysine methyl can silence or activate the gene

*generally methyl addition is usually linked with silencing of a gene or even whole chromosome
* histone methylation is responsible for switching off one X chromosome in every female cell to produce Barr body

Stem cells

Zygote → morula → blastocyst → fetus

Morula: solid ball of totipotent stem cells
[totipotent: can differentiate to all types of cells]

Blastocyst: outer cells have differentiated to form placenta
Inner cells are pluripotent stem cells
[can differentiate to any cell but placenta cell]

Stem cell source:

1) embryonic stem cells
Has plenty of Totipotent stem cells
Ethical issues: can be illegal/ can develop to cancer/ risk of rejection/ can cause damage to baby
Yet is the easiest to get and has best stem cell type

2) umbilical cord
Blood drained from it has plenty of pluripotent stem cells
Ethical issues: less availability/ can develop to cancer/ risk of rejection
Pros: no damage to human

3) adult stem cells (somatic stem cells)
Has multipotent stem cells
[multipotent: are found in specialized tissues but are undifferentiated cells. Such as bone marrow cells that make blood cells]
Ethical issues: as age increases number of stem cells decreases / can develop to cancer/ risk of rejection/ difficult to extract / form a limited range of differentiated cells
Pros: no damage to human

4) induced pluripotent stem cells (iPS)
Reprogrammed adult cells to give pluripotent stem cells
Ethical issues: not easy to make cells pluripotent/ differentiating them again difficult/ high risk of cancer/ risk of cell to differentiate back to original adult cell
Pros: no risk of rejection/ no damage to human

Treatments using stem cells

- stem cell therapy

cells are taken from embryo are synthesized
cells are cultured and returned to patient
was used successfully to form new organs

- therapeutic cloning

cells are taken from patient and nucleus is removed
nucleus is moved into an human ovum (ovum nucleus is removed)
mild electric shock is used to fuse the new cell and trigger development
new cell divides by mitosis to produce genetically identical cells
stem cells are harvested and cultured to differentiate to required cell
when cells develop into tissue they are transferred to patient

Treating conditions such as:

1) Parkinson's Disease

Nerve cells stop producing dopamine which causes body to stop moving
Stem cells from mice are cultured to from dopamine neurons which are transferred to patient; production of dopamine rises again and patient can control body movement better

2) Type 1 Diabetes

Islet of langerhans cells are damaged and can't produce insulin so glucose blood level is not controlled
If islet cells can be developed from stem cells then patients could completely discard of insulin injections and diet monitoring

3) Damaged Nerves

If Damages and destroyed nerve cells in brain and spine are replaced by stem cells then control of damaged area can be regained

4) organ transplant

Stem cells are cultured to from needed tissues from stem cells and are transplanted to patient to grow into organs. If cells cultured are from patient then there is no risk of rejection

Ethical questions

- respect of autonomy → not preforming procedures without consent
- beneficence → aim is doing good; to relieve suffering
- non-maleficence → doing no harm
- justice → treating everyone equally and sharing all resources fairly with no discrimination

Issues revolving around use of stem cells

-many disagree with the use of embroynic stem cells as it damages the embryo which is a human with the right to live

-relatively small number who are willing to be egg donors
-adult stem are limited to the types of cell they can differentiate into

-storing pluripotent stem cells from blood drained from umbilical cord is very difficult and expensive

Regulatory authorities and stem research

1) monitor research and ensure it's necessary
2) issue licenses
3)monitor sources of stem cells
4)ensure that only early stage embryos are used
5)prevent unethical use of stem cells such as human cloning

Phenotype expression

Phenotype -> genotype + environment

- genotype: combination of alleles

Multiple alleles:
A certain phenotype depends on multiple alleles one from each parent
-An allele in combination can be: dominant, recessive or codominant

Discontinuous Variation within a species is usually a phenotype mainly dependent on genotype

-Blood type: depends entirely on genotype, alleles passed down from parent

-Has no range only specific groups
*Discontinuous variation is represented by bar graph

Gene linkage:
When genes are closer to each other on chromosome, percentage of being inherited together increases

Polygenic inheritance:
When a trait is determined by more than one gene

Each allele in combination Has an effect on phenotype produced
-Recessive genes are not totally masked by dominant, they get expressed at different percentages depending on their number in a combination

Creates continuous variation in a species as trait will differ (slightly) if one allele changes

Continuous variation within a species is due to phenotype being affected by both environment and genotype

-Height: due. To nutrition and genes inherited from parents

-Has a range of phenotypes
*represented by normal distribution graph and histogram

- environment: interactions with biotic and abiotic factors in a habitat

A) height → polygenic trait
Depends on nutrition an individual intakes
Tall Parents may hav a shorter child if child has malnutrition/ calcium deficiency or a disease which effects growth

B) cancer → heavily environmental
Mutation in gene causes tumor suppressor to not be produced/ damaged so cell cycle is not controlled
Mutation is usually from environmental reasons such as:
UV exposure —> skin cancer
Tar inhalation (smoking)—> lung cancer

C) Siamese cats → due to genotype and environment
Genotype: gene coding for tyrosinase enzyme
[which coverts tyrosine to melatonin]
Gene coding for enzyme gets mutated as a result new enzyme formed is only active at low temperatures
Environment: temperature, only coldest parts of cat get active tyrosinase and produce the colour pigment, melatonin
(Coldest parts in mammal are their tips)

Topic four

Plant cells

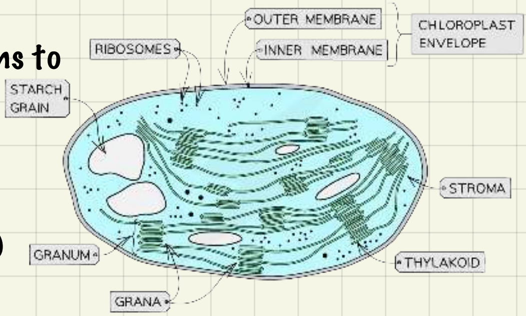
Are eukaryotic cells which have all the organelles found in animal cells except for centrioles

Organelles found in plants but not animals:

1) chloroplast

Are found in the green part of most plants

- have folded inner membrane for increased surface area for reactions to take place
- Have an outer membrane
- Contain it's own DNA
- Has 70s ribsome
- **Have chlorophyll (green pigment) which traps energy from sun for photosynthesis**



2) amyloplast

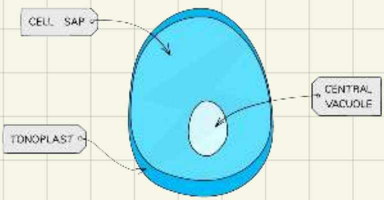
Starch storage unit

- starch is made from alpha glucose polymers, amylose + amylopectin
- Can convert starch into glucose when energy is needed

3) permanent vacuole

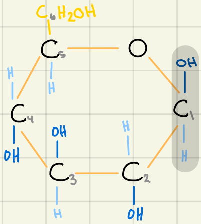
Fluid-filled space surrounded by a membrane

- has a specialized membrane, **tonoplast**, which contain protein channels and carriers system to control movement of molecules into/out of vacuole so it controls water potential of cell
- Contains cell sap, sugary solution, which makes water move into cell by osmosis maintaining cell pressure and shape
- Acts as a store, can store pigments/ waste/ proteins



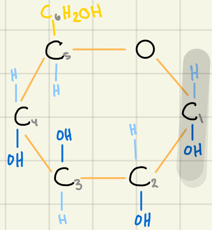
Chemistry of cellulose

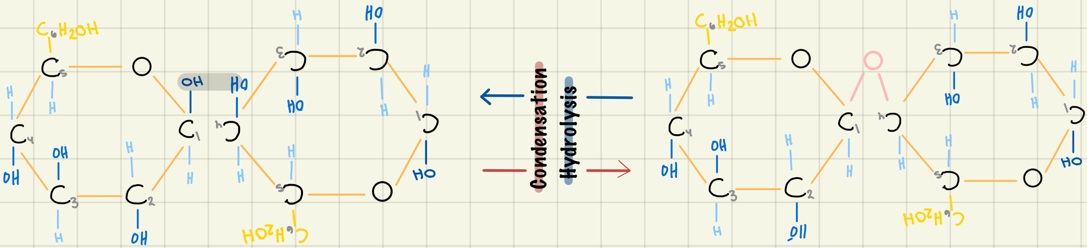
- **Beta glucose:**



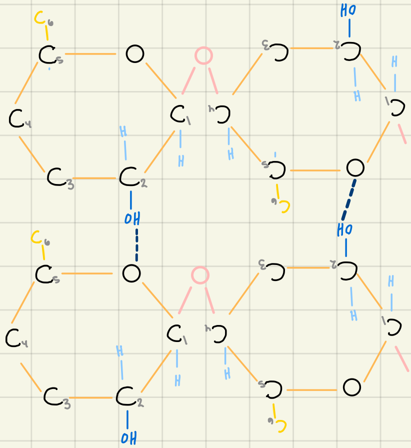
Only difference is positions of atoms in side chain C1

* alpha glucose





Only form 1,4 glycosidic bond and hydrogen bonds; is straight chain



Hydrogen bonds

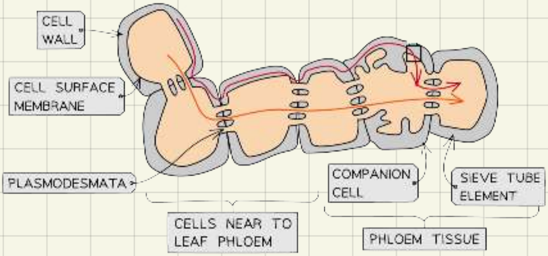
Structure of cellulose microfiber

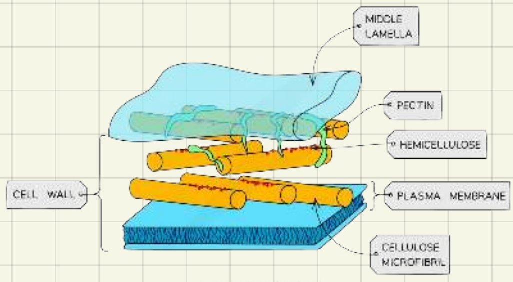
4) cell wall
Hard membrane which gives plant cells their regular shape
- has plenty of hydrogen bonds for strength
- has lignin to decrease permeability and add strength
- Made from insoluble cellulose

—> primary cell wall (in young plants)
Cellulose microfibers are parallel to each other making the cell wall very flexible

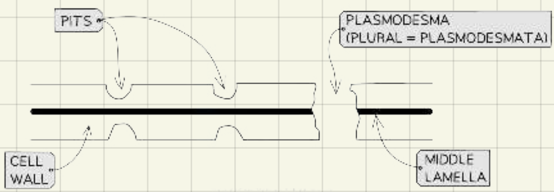
—> secondary cell wall (in mature plants)
Through a process called secondary thickening cellulose microfibers are densified and have an over-lapping arrangement
+ lignin (dead tissue) is added to strength structure
* primary structure is still found over it

- middle lamella is first layer and it joins to cells together
has pectin (carbohydrate) + calcium pectate for a sticking effect
Is made when a cell divides into two new cells





5) plasmodesmata
Cytoplasmic bridge
- allows communication between cells
- Cells don't separate completely after cell division
- At plasmodesmata region cell wall is thin, this region is called pits
- Pits allow water to move between different xylem vessels



Plant stem

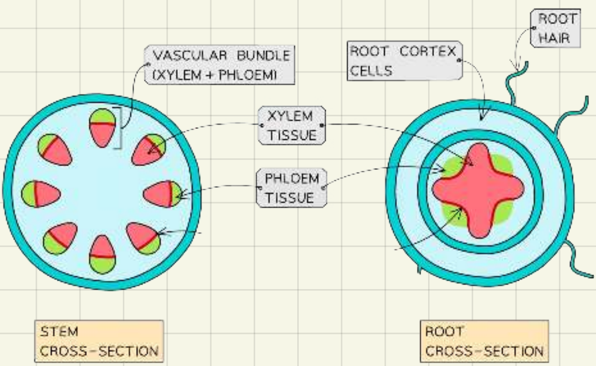
__Stem function__
1) holds plant upright and supports it
2) act as transport system for water and nutrients

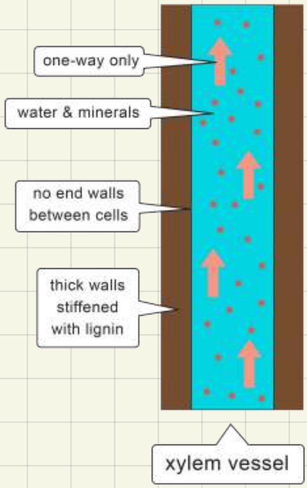
__Tissues in stem include:__
1) xylem
Takes in water and minerals from roots and transport it to photosynthetic cells
* only transports in one direction, upward.

—> in young plants: xylem is still made up of living cells called protoxylem
not lignified yet so can stretch and grow

—> when stem matures: cells stop growing and internal content dies, metaxylem
Lignin is added; xylem vessel is a whole continuous tube
* lignin makes xylem water proof/ prevents transpiration

- xylem vessel is made of dead, hollow cells on top of each other with no cell wall between them to create a continuous path for water and minerals

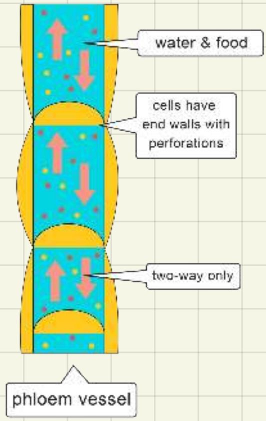




2) phloem

Transports sucrose/ amino acids from source to cell in an active process called translocation

* moves in both directions



Is made up of many living cells joined together to make a tube

- have sieves (holes) so phloem sap can moves through tube
- Mature Phloem cells have no nucleus and survive due to companion tissues

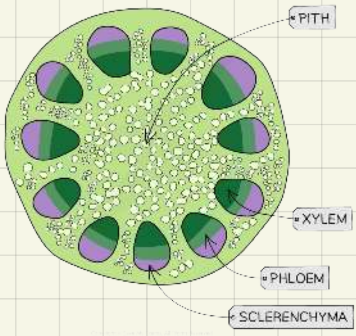
* companion tissues are normal active cells that connect to phloem by plasmodesmata to provide it with needed nutrients and energy for active transport

3) sclerenchyma

Made from dead cells, secondary cellulose and is lignified

- since it is lignified water can't enter cells and they die
- supports the plant
- Is not involved in transportation since the ends of cells are tapered (not open)

* is very similar to xylem only difference that sclerenchyma is not part of the plant transport system



Cross section from stem

Needed minerals in plants

1) magnesium → production of chlorophyll for photosynthesis
2) nitrates → production of amino acids which produce protiens/ enzymes for growth and repair
3) calcium → production of calcium pectate which combines with pectin to hold plant cells together (middle lamella)

Investigation to show effect of deficiency of a named mineral

- seedlings/shoots of same age and species and measure initial height
- put in soil pots of same soil pH
- keep under constant conditions (temperature, humidity, light intensity)
- one of the pots should be a control and have all minerals
- each other pot will have one missing mineral
  * all should be added in same volume and concentration
- after a period of time measure final height
- measure difference and repeat test several times to get mean

^find optimum concentration of Mg needed for optimum growth.
Same steps except all should contain all minerals but each pot will have a different concentration of Mg

Uses of plants

A. As a source of Food

B. Plant fibers

Extracted from plant, sclerenchyma and xylem tissue

How to extract sclerenchyma and xylem from plant?

-Cut stem and soak in water or chemicals to soften living tissues

-Add enzymes which will remove living tissues

(Or by retting: leave it till natural decomposers break down living tissues)

-Since xylem and sclerenchyma are dead they will be left behind

[plant such cotton do not need the extraction process]

- Cellulose and lignified cellulose make plant fibers:
  - not easily broken down by chemicals/ enzymes
  - have great tensile strength

Investigation : how to measure tensile strength

- take same length, width and diameter of plant fiber
- with same pretreatment [soaking in water]
- fix end of fibre with a clamp
- add small masses gradually until fibre breaks

*to find force needed to break fibre → mass/ cross sectional area

- repeat investigation for each fibre type and get mean

⇒ plant fibre can be made into clothes, ropes

→ synthetic fibers (Nelon/polystyrene)

Made from oil/ fossil foils

Pros: cheaper, can be made into different designs

Cons: doesn't absorb water, not sustainable, creates green house gasses

→ bio-plastic

From polymers of starch in plants

Pros: sustainable, biodegradable, reduces pollution

Cons: more expensive than oil based plastics

Uses: medication capsules, computer casting, drinking cups, bank notes

C. Wood

Made of composite material (lignified cellulose)
which makes wood:

- strong and can support large weights
- flexible doesn't crack easily
- good insulator

*is carbon neutral, taking in carbon when growing and releasing it when burnt
Is recyclable; sustainable

Uses → paper production
→ furniture
→ construction

D. Medicine

Plants have antimicrobial properties → has ability to kill bacteria and fungi
* active ingredient is extracted to make medicine

Advantages of extraction:

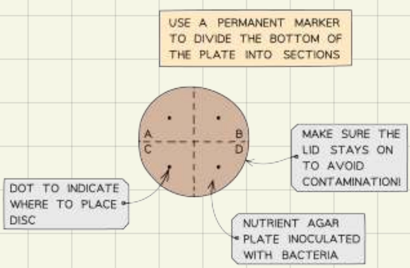
- give suitable, repeatable doses of active ingredient
  *at different seasons same plant may have different concentrations of a certain substance
- purifying drug from any unwanted substance

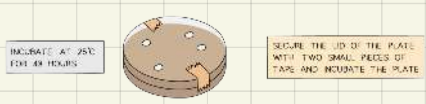
⇒ first person to introduce concept of active ingredient
↳ William weathering
Treated patients with heart condition with a plant he found in digitalis soup
He experimented with different concentrations of plant until he found the correct dose and in the way noted the side effects
The Plant had the active ingredient which is now the drug digoxin

Design an investigation to show that garlic has an antimicoubial effect

- Get extract from garlic by crushing then dissolve in solvent and get filtrate. under aseptic (sterilized) conditions
- Put agar in sterilized Petri dish
- Spread non-pathogenic bacteria on surface of agar
- Add garlic extract by:
    1) making holes/wells of same size and adding equal volume and concentration of extract
    2) soak filter paper discs of same size in extract then place discs on surface of agar
- As a control add water to one of the wells/ filter paper discs
- Cover plate, without sealing completely as it prevents aerobic respiration,
- incubate at 25C [37C will encourage growth of pathogenic bacteria] for 48 hours
- Clear zones/ zone of inhibition: antibacterial enzymes from extract diffused so no bacterial growth is recorded
- Measure diameter of inhibition zone to see effectiveness

- Agar is a substance used for culturing bacteria it has nutrients needed by bacteria
- Petri dish + agar = plate
- Plate + incubator provides all ideal conditions for bacterial growth





Developing new medicines

New drugs must be:

- effective → cures, prevent or relieves symptoms of a disease
- Safe → non-toxic with no substantial side-effect
- Stable → can be stored and used under normal conditions
- Easily taken into and removed from body
- can be made on a large scale → can be produced in pure form, in large quantities and cheaply

Stages of drug development:

- pre-clinical

  Extraction of active ingredient and understanding how it works
  Is first tried on cells then tissues to measure effectiveness
  If effective then it is tried on animals

⇒ animal testing and drug development

- expensive, time consuming and surrounded with ethical issues
  But testing on cultured cells alone is not sufficient
  And they are not used for pleasure but to ensure human safety

- clinical trials

  ① small group of healthy volunteers are given the new drug to see side effects on human body
  ^ some are given placebo as a control

  ② small group of patients (volunteers)with target disease are split into 2 groups. → one is given new drug
  → second is given a placebo to eliminate psychological factors

  ③ large group of patients (volunteers) is treated with new drug and placebo and/or an existing drug to compare effectiveness
  Larger group is used to see different side effects in different bodies

Double-blind trials

*in both rials II and III double-blind trials method is used

Patients are divided into 3 groups

→ 1) takes new drug

2) takes the best available drug

3) takes a placebo

Neither the doctors nor the patients know what the patients are taking

This is to eliminate psychological effect and bias

To see if there is significant difference between the two drugs

→ It is difficult to give accurate results in clinical trials since many patients stop taking the medication for various reasons or some do not take it regularly

→ If drug is found very successful then it is unethical to deny real medication for patients taking placebo or other drugs

→ Even if medication was successful in all stages it still need to licensed before it can be put in market and after it is put in market, research continues using feedback from patients who bought it to note long term side-effects

Bacterial growth

Reproduce by binary fission. in ideal conditions reproduction is very quick

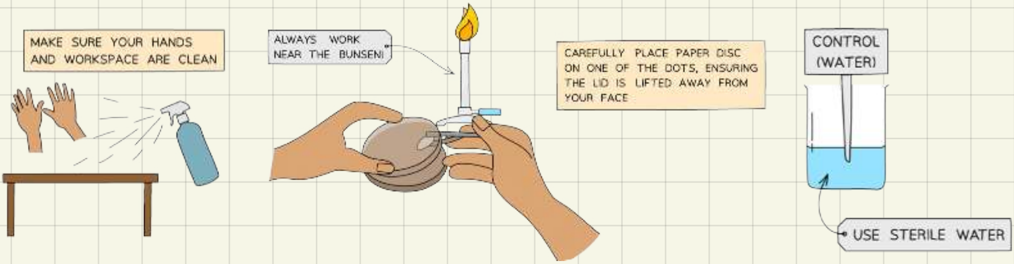
What are the ideal conditions and why are they needed?

- optimum pH and temperature → optimum enzyme activity
- oxygen → aerobic respiration for energy
- water → as a solvent
- nutrients (amino acids and sugars)→ for protein production
  → for aerobic respiration

Bacteria is cultured in Petri dish which has agar under aseptic condition

- aseptic conditions:
  → all equipments must be sterile
  → leave an open Bunsen burner to kill airborne bacteria
  → equipments must be sterilized between uses
  → wipe benches with alcohol
  → used equipments must stay in lab and be disposed of correctly

- Aseptic condition ensure:
  → safety of workers, that they are not infected by pathogenic Bacteria accidentally and spreading infection to other people,animals,plants
  → That no cross-contamination occurs
  → Less competition between different types of bacteria and wanted bacteria



Classification

Due to evolution variation between organisms intensified
So to be able to study them they had to be sorted

Taxonomy: the science of describing, classifying and naming organisms

Organisms are classified depending on similarities and differences in their morphology (physical appearance)

Taxonomic groups:
Kingdom, phylum, class, order, family, genus and species

The 5 kingdoms are
1) Eubacteria
2) Protista
3) Fungi
4) Plantae
5) Animalia

→ a new classification: molecular phylogeny
Some organisms have similar morphology but different genetic material and vice versa
So DNA analysis to study DNA base sequence, amino acids and proteins is used

New taxonomic group: domain

The 3 domains are
1) Eukaryota (animal, plants, fungi)
2) Bactria
3) Archea

* Binomial naming system: universal name for an organism
  - made of 2 parts
    1. Genus, 1 letter is capital
    2. Species, small letter
  * can be written in italic

Models for classifying species

1) Reproductive Species Model
Species: closely related individuals that can mate and produce fertile offsprings
A group of organisms which genes can flow between individuals

Limitation: Bactria produces A-sexually and doesn't mate
Some different species mate to produce fertile offsprings

2) Ecological Species Model
Species: individuals with similar niche (role in ecosystem)

Limitation: roles differ as ecosystem differ and some organisms may have occupy different roles

4) Mate-recognition Species Model
Species: individuals with similar mating behavior

Limitation: cross pollination can occur between different species
Fossil organisms can't reproduce

5) Genetic Species Mode:
Species: individuals with very similar DNA sequence

Limitation: no decided max difference in genes until individuals are from different species
New methods, still expensive
Fossil organisms don't have accessible DNA

- **Evaluating new studies**

Scientific data is published in scientific journal or conferences

Supports data with evidence

Peer review, other scientists revise evidence and analyze data

More scientists collect their own new evidence to:

→ Accept new study

→ reject new study

→ accept with modifications

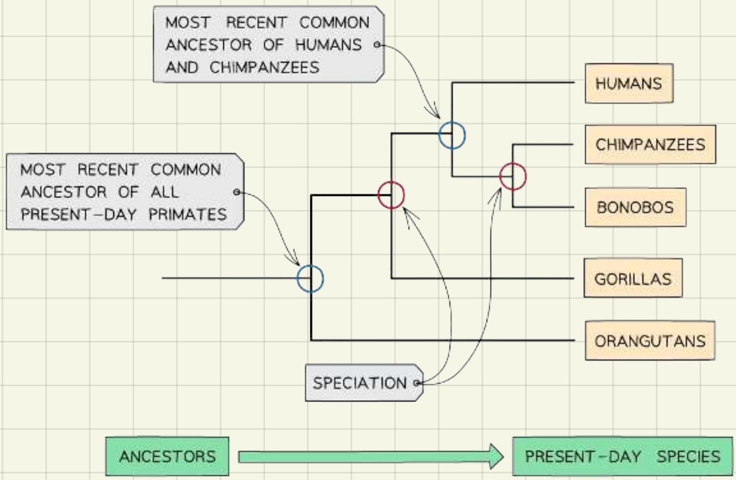
-**DNA sequencing**: when all or part of an organisms genome is worked out

-**DNA profiling**: when only the non-coding areas of an organisms genome is worked out

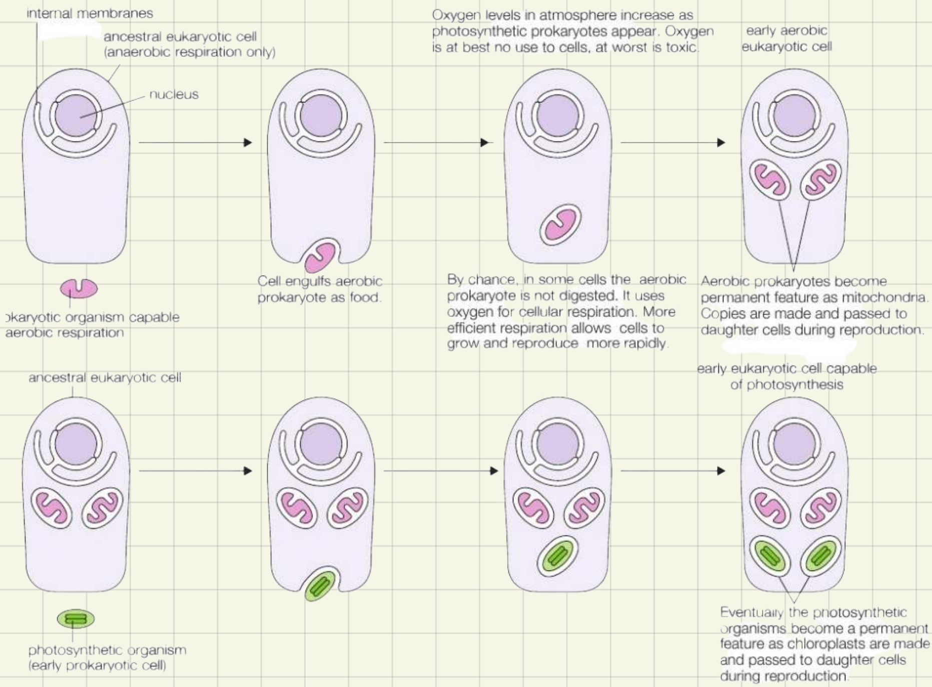
*Both can show pattern differences and similarities between individuals

-Is used for: classifying organisms, finding paternity, identifying criminals

Phylogenic tree: show relation between different organisms



Model of how eukaryotic cells may have developed



Characteristics	Bacteria.	Archaea.	Eukaryota
Membrane enclosed Nucleus	Absent	Absent	Present
Membrane enclosed Organelles	Absent	Absent	Present
Peptidoglycan in Cell wall	Present	Absent	Absent
Ribsome type	70s	70s	80s
RNA polymerase	1	1	3
Some conduct chlorophyll Based photosynthesis	Yes	No	Yes

Biodiversity in ecosystem

Biodiversity: variation between species and individuals in a species

All organisms in an ecosystem are interdependent and all ecosystems are interlinked
So a change (extension) in one species will effect whole ecosystem(s)

High biodiversity helps maintain ecosystems as individuals are able to adapt and survive

Factors increasing biodiversity

→ stability in an environment (no sharp changes in temperature, rainfall)

→ larger areas and availability of nutrients decreasing competition

→ high reproductive rates as it results in higher (good) mutation rate increasing variation

→ high level of productivity (high rate of photosynthesis)

* rainforest has high biodiversity as it has stable warm temperature with Large volumes of rainfall and has plenty of nutrients and large areas so Rate of reproduction of species is high

Endemism: when a certain species is only found in one area
- risk: interbreeding so genetic variation drops and specie can't Easily adapt and survive in a changing environment

Measuring biodiversity:

* no specific time/ season to measure biodiversity
  But after conducting first measuring all next annual.
  measurements should be in same time/seasons

Levels of biodiversity:

1) at a species level

   i) species richness: number of species found in an area

- diversity index $D = \frac{N(N-1)}{\Sigma n(n-1)}$

  D= diversity index
  N= Σ Number of organisms of all species
  n= Σ Number of organisms in each species

   ii) relative species abundance: relative number of organisms in a species

- Quadrat for measuring species richness/abundance

Used to make counting and sampling individuals easier and more accurate.

Should be placed at random areas on several positions

2) within a species (genetic biodiversity)

- Allele frequency : frequency of a certain allele appearing within a species
  ⇒ affected by:
  i) mutation
  → producing an advantageous allele
  So allele stays in population and allele frequency increases
  → producing disadvantageous allele
  So allele is removed and allele frequency decreases

* a disadvantageous allele in an environment may be an advantageous gene in an other environment

  ii) migration

- measuring genetic biodiversity (level of difference in genet make-up)

$$\text{- heterozygosity index} = \frac{\text{number of heterozygotes}}{\text{number of individuals in population}}$$

*The higher heterozygosity index the higher the genetic diversity

Adaptations to a niche

Niche: role of an organism in a community

Adaptation ensures survival of individuals and passing down their genes

Types of adaptations

1) anatomical adaptation
   Form and structure of an organism
   → Thick layer of blubber in seals and whales to protect from the cold

2) physiological adaptation
   The work of an organism's body (metabolic reactions, enzymes)
   → vasodilation and camouflage

3) behavioral adaptation
   Programmed/instinctive activities
   → emigration and hibernation

Gene pool: sum of all genes in a population at a given time

Allele frequency in a balanced population follows Hardy- Weinberg equilibrium

A balanced population doesn't have:
—> Selection pressure, all alleles good and bad will be passed down equally through generations
—> Mutation; no new alleles are introduced by sudden mutation
—> Migration (isolated population) no alleles are leaving or entering population
—> Small population size, easier to maintain useful alleles than in larger populations
—> Non-random mating

1) $p^2 + 2pq + q^2 = 1$

2) $p + q = 1$

p —> number of dominant homozygous
pq —> number of heterozygous
q —> number of recessive homozygous

The easiest to calculate is q since heterozygous and dominant homozygous share same phenotype
[question will give $(\text{probability})^2$]

In a real life there so no balanced population as multiple factors can cause increases or decrease in allele frequency such as:
-selection pressure
-mutation
-random mating
-migration

If sum of p2 + 2pq + q2 = 1
> 1 then allele frequency increased
< 1 then allele frequency decreased

Speciation

↳ Formation of new species due to isolation

Types of isolation

1) geographical isolation (allopatric speciation)
Is when population become physically isolated as when islands split

— one population is split due to environmental changes so each part of population is exposed to a different environment and different selection pressure
So each new population will have different advantageous alleles to pass down to their offsprings after a while each population will have a different gene sequence producing 2 new species that can't mate to produce fertile offspring

2) hybridization
Closely related species crops breed to produce fertile offsprings which may not be able to cross back with parent species and are better adapted to the niche out-competing their parents

3) ecological isolation:
2 populations inhibit the same region but each type prefers different type of habitat

4) seasonal isolation:
sexual receptiveness in some part of a population becomes different than the rest of th population

5) behavioral isolation:
change in the mating pattern so some individuals can't recognize others as being potential mates

6) mechanical isolation:
mutation to the reproductive system of organisms so they can only mate with some members of the population

* reproductive isolation (sympatric speciation) is the most important factor in isolation

Natural selection

Only individuals that can adapt to a change in the environment (heavy rain) will survive and reproduce.
So advantageous allele flows through all next generations ones that could not adapt will die and their allele is not passed down
So allele frequency changes
↳ evolution, since disadvantages allele is moved from population and all new individuals have advantageous allele

How allele frequency change leads to evolution?

In large population there is variation up to mutation
Under certain selection pressure the individuals whom have advantageous allele will adapt and survive
Most of other individuals will die as they can't adapt since they don't carry advantageous allele
Only individuals that can adapt and survive will mate and pass down their advantages allele to their offsprings
By the time the allele passes through generations this leads to a change in allele frequency (evolution)

Adaptive radiation

After speciation, up to selection pressure only individuals with wanted features survive as they adapt to environment and reproduce
Other unadapted individuals will die removing their features from ecosystem

Darwin's finches

Charles Darwin observed a group of small sparrow-like black birds with strong, short beaks. These finches varied on different islands, but they were closely related to one another. He observed those finches and concluded that they came from a common ancestry that had previously migrated from the mainland to the volcanic islands and had undergone changes to adapt to the different conditions of the individual island. The different geographical landscapes in the Galapagos Islands have promoted the diversification of finches on the same island as each region had different selection pressure so it was important to mate with similarly shaped shaped peak to pass down on the advantageous allele.

Population bottlenecks

A large population is needed to maintain a large and diverse gene pool
Due to some event (diseases, climate change, over hunting)
Population size decreases → allele frequency decreases → gene pool decreases → variation (genetic diversity) deceases
So new (smaller) population is very vulnerable to changes in environment

* <u>causes limited gene pool due to death of large number of members from natural events or human intervention</u>

Founder effect

Loss of genetic variation that occur when a small number of individuals leave the main population and see up a new population
(voluntary population bottleneck)
New group will most probably not have all alleles found in main group so any unusual genes in founder of new group will become more common as their population grows creating a significantly different gene pool than original population

* <u>causes limited gene pool due to isolation of small group from original population</u>

Conservation

Human population explosion

Due to increased awareness, education and development in medicine and other industries humans were able to provide for themselves with food, shelter and protection efficiently

Though a lot of the activities that help us be this efficient are destroying the ecosystem and causing loss of biodiversity such as
-Habit destruction to grow crops or build houses which causes extinction of some species
-Climate change which causes an increasing number of extreme weather in a short period so lots of species won't be able to adapt and survive
-Depletion of biological resources as human demand for food and a great variety of it increases plenty of lands and resources are going to be used to meet it causing lack of nutrients and shelter for other species
*All these actions lower biodiversity on both species and genetic level

Conservation: keeping and protecting a living and changing environment
<u>Ways of conservation</u>:
1) reclaiming land after industrial use
2) setting up sustainable agriculture systems
3) protection of threatened species
4) global legislation on pollution levels and greenhouse gas emissions

- **Programs to help threatened species**
  - in-situ conservation (in their habitat)
    - ↳ monitor population size to see it they are under the danger of extension
    - → Help them survive by providing nutrients, shelter, protection

* not always successful and hard to maintain

  - ex-situ conservation (out of habitat)
    - ↳ in zoos , farms
    - → Used when species is on verge of extinction and too late for in-situ conservation to at least store genetic material
      1) seed banks
      2) captive breeding programs

1) seed banks

Seeds are used since they
- have diploid genetic material
- don't effect population size
- have no ethical concerns/issues
- are small, can be easily stored

Process:

→ collect large number of seeds from different plants from same species (To increase genetic variation)

→ X-ray to see if embryo is present and not mutated or damaged (can germinate)

→ wash and sterilize (so bacteria won't decay it)

→ store at low temperature (deactivate enzymes so seed doesn't decompose or germinate)

→ store at low humidity (doesn't decompose)

→ annual check for viability by growing a sample if more than 70% germinates then process is successful

Some plants can't be conserved this way so either field gene banks are used which keep growing the plant year after year (takes huge areas) Or tissues culturing to conserve plants genetic material (takes less space and helps more variety to be conserved)

2) captive breeding programs

Held in zoos and wildlife parks as they offer:

- protection from predators
- nutrients
- shelter (imitated from their habitat)

→After population size regrows they are reintroduced to original habitat

* doesn't always work but can be put in protected areas or national parks

Issues:

Expensive and time consuming
Difficult to provide right conditions
Reintroduction doesn't always work
Large areas are needed
Small group (low genetic variation)

↳ to help with this issue, cross-breeding with other zoos (IVF) to increase genetic diversity
As well as stud books are used to keep track of of genetic data for their breeding individuals

- Increasing Sustainability by:
  - -Selective filling for deforestation
  - -Avoiding monoculture
  - -Using organic fertilizers and lower use of fertilizers
  - -Using biological pests instead of pesticides
  - -Less use of machinery (consume lots of fossil fuels)
  - -Maintaining sustainable agriculture and tourism

* All of this helps loss of biodiversity in the first place

- Awareness and education

of damage and impact of human activities on ecosystems
And understanding that ecosystems are interdependent on each other
And knowing ways of protecting and supporting species and their habitat